International Journal of Medicinal Mushrooms, 15(6): 539–554 (2013)

Neurotrophic Properties of the Lion's Mane Medicinal Mushroom, *Hericium erinaceus* (Higher Basidiomycetes) from Malaysia

Puei-Lene Lai,[1,4] Murali Naidu,[2,4] Vikineswary Sabaratnam,*[1,4] Kah Hui Wong,[1,4] Rosie Pamela David,[2,4] Umah Rani Kuppusamy,[3,4] Noorlidah Abdullah,[1,4] & Sri Nurestri A. Malek[1,4]

[1]Institute of Biological Sciences, Faculty of Science, University of Malaya, Kuala Lumpur, Malaysia; [2]Department of Anatomy, Faculty of Medicine, University of Malaya, Kuala Lumpur, Malaysia; [3]Department of Molecular Medicine, Faculty of Medicine, University of Malaya, Kuala Lumpur, Malaysia; [4]Mushroom Research Centre, Fungal Biotechnology Lab, Institute of Biological Sciences, University of Malaya, Kuala Lumpur, Malaysia

*Address all correspondence to: Vikineswary Sabaratnam, Institute of Biological Sciences, Faculty of Science, University of Malaya, 50603, Kuala Lumpur, Malaysia; Tel.: 603-7967 4349; Fax: 603-7967-4178; E-mail: viki@um.edu.my

ABSTRACT: Neurotrophic factors are important in promoting the growth and differentiation of neurons. Nerve growth factor (NGF) is essential for the maintenance of the basal forebrain cholinergic system. Hericenones and erinacines isolated from the medicinal mushroom *Hericium erinaceus* can induce NGF synthesis in nerve cells. In this study, we evaluated the synergistic interaction between *H. erinaceus* aqueous extract and exogenous NGF on the neurite outgrowth stimulation of neuroblastoma-glioma cell NG108-15. The neuroprotective effect of the mushroom extract toward oxidative stress was also studied. Aqueous extract of *H. erinaceus* was shown to be non-cytotoxic to human lung fibroblast MRC-5 and NG108-15 cells. The combination of 10 ng/mL NGF with 1 µg/mL mushroom extract yielded the highest percentage increase of 60.6% neurite outgrowth. The extract contained neuroactive compounds that induced the secretion of extracellular NGF in NG108-15 cells, thereby promoting neurite outgrowth activity. However, the *H. erinaceus* extract failed to protect NG108-15 cells subjected to oxidative stress when applied in pre-treatment and co-treatment modes. In conclusion, the aqueous extract of *H. erinaceus* contained neuroactive compounds which induced NGF-synthesis and promoted neurite outgrowth in NG108-15 cells. The extract also enhanced the neurite outgrowth stimulation activity of NGF when applied in combination. The aqueous preparation of *H. erinaceus* had neurotrophic but not neuroprotective activities.

KEY WORDS: medicinal mushrooms, *Hericium erinaceus* combination, cytotoxicity, enhancement, extracellular NGF, medicinal mushroom, neurite outgrowth, neurofilament, oxidative stress

ABBREVIATIONS: bFGF, basic fibroblast growth factor; **BDNF**, brain-derived neurotrophic factor; **DMSO**, dimethyl sulfoxide; **GDNF**, glial cell line-derived neurotrophic factor; **H_2O_2**, hydrogen peroxide; **MTT**, 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyl-tetrazolium bromide; **NGF**, nerve growth factor; **NT-3**, neurotrophin-3; **PBS**, phosphate-buffered saline; **RMP,** resting membrane potential; **TUNEL**: terminal deoxynucleotidyl transferase-mediated dUTP nick-end labelling

I. INTRODUCTION

The formation of axons and dendrites (collectively known as neurites) from individual neurons is pertinent to the functionality and development of nervous system. Neural connectivity relies on the cellular mechanisms of axon outgrowth and synapse formation. Neurotrophic factors are critical for the survival, function, and connectivity of neurons. An adequate level of NGF is important for the maintenance of a functional basal forebrain cholinergic system, and this dependence of trophic support is augmented in aging brains.[1] Being a potent survival factor for cholinergic neurons, NGF has been extensively studied for its therapeutic effect in the treatment of Alzheimer's disease.

The Lion's Mane medicinal mushroom, *Hericium erinaceus* (Bull.:Fr.) Pers. (Hericiaceae, higher Basidiomycetes) is an edible mushroom widely consumed in the Orient and highly reputed for its medicinal values.[2] The discovery of NGF-stimulating compounds in this mushroom has attracted much

1045-4403/13/$35.00 © 2013 Begell House, Inc. www.begellhouse.com

attention in research, as it holds promise as a cure for Alzheimer's disease. Phenol derivatives, hericenones C-H[3,4] were isolated from fruiting bodies, while erinacines A-H[5-8] were isolated from mycelium. These compounds accelerate the synthesis of NGF and stimulate neurons to regrow. Furthermore, dilinoleoyl-phosphatidylethanolamine (DLPE) isolated from *H. erinaceus* has been shown to protect neuronal cells from endoplasmic reticulum (ER) stress-dependent cell death.[9] A double-blind, placebo-controlled clinical trial showed signs of improvement in subjects with cognitive impairment after consumption of *H. erinaceus*.[10]

Recent studies have explored the pharmacological properties of two or more neurotrophic factors, when used together, to promote neuronal survival and regeneration. Axonal growth guidance in embryonic lumbar dorsal root ganglion cells (DRG) was enhanced when NGF and NT-3 were used in combination.[11] The combinatorial administration of bFGF, NT-3, and BDNF synergistically enhanced neuronal survival, disinhibited axon growth, and promoted axon regeneration without the intervention of scar tissue at the lesion site in retinal ganglion cells.[12] A recent study[13] has shown that the combination of GDNF and NGF works synergetically in the process of axonal elongation, axonal branching, and growth kinetics.

Hericium erinaceus cultivation has been successfully adapted to the tropical climate in Malaysia, and information on this locally adapted species is limited. Previous studies have shown that the mushroom retained its neurite-stimulating activity[14] and enhanced peripheral nerve regeneration *in vivo*.[15] This medicinal mushroom has great potential to be developed as a functional food or nutraceutical for boosting brain and nerve health. Aqueous preparation was used in this study to simulate the cooking process. This preparation method also breaks down the cell walls of mushroom, allowing the medicinal components to become available and thereby readily absorbed once consumed.[16] In this study, we evaluated the synergistic interaction of the aqueous extract of Malaysian-grown *H. erinaceus* and exogenous NGF on the neurite outgrowth stimulation activity in NG108-15 cells. The neuroprotective effect of the mushroom extract toward neuronal cells under oxidative stress was also studied.

II. MATERIALS AND METHODS

A. Preparation of Mushroom Extract

The mushroom extract was prepared according to the method of Wong,[14] with slight modifications. Briefly, fresh fruiting bodies of *H. erinaceus* obtained from Vita Agrotech Sdn. Bhd. (a mushroom farm in Tanjung Sepat, Selangor, Malaysia) were freeze-dried at –50 ± 2°C for 48 h and ground into powder. The mushroom powder was soaked in distilled water 1:10 (w/v) overnight, boiled for 30 min with agitation, and filtered. The filtrate was freeze-dried and stored at -20°C. Prior to assay, the freeze-dried filtrate was dissolved in distilled water to the required concentration and sterilized with a micropore filter of size 0.2 μm.

B. Cell Culture

The neuroblastoma-glioma hybrid NG108-15 was purchased from ATCC (American Type Culture Collection, USA) and maintained in Dulbecco's Modified Eagle's Medium (DMEM, Sigma-Aldrich, St. Louis, MO, USA), supplemented with 10% heat-inactivated fetal bovine serum (FBS, PAA Lab GmbH, Austria), 100 U/mL penicillin/streptomycin (PAA Lab GmbH, Austria), 100 μM hypoxanthine, 0.4 μM aminopterine and 16 μM thymidine (HAT, Sigma-Aldrich).[17] The human lung fibroblast MRC-5 was grown in Eagle's Minimum Essential Medium (EMEM, Sigma-Aldrich) supplemented with 10% heat-inactivated fetal bovine serum (FBS, PAA Lab GmbH), 1 mM sodium pyruvate (Sigma-Aldrich), 1.5 g/L sodium bicarbonate (Merck, Darmstadt, Germany), 100 U/mL penicillin/streptomycin (PAA Lab GmbH, Austria) and 50 μg/mL of amphotericin B (PAA Lab GmbH). The cell lines were cultured at 37°C in a 5% CO_2 humidified incubator (Shel Lab, Oregon, USA). The medium was changed every 2–3 days as needed and cultured to achieve at least 70% con-

fluence prior to assay.

C. Assessment of Cytotoxicity of *H. erinaceus* Extract

The toxicity of *H. erinaceus* aqueous extract on NG108-15 cells and human lung fibroblast, MRC-5 was tested using MTT assay. The cells were seeded in 96-well microtiter plates (Orange-Scientific, Braine-'Alleud, Belgium) at a seeding density of 5×10^4 cells/mL and incubated overnight. Various concentrations of mushroom extract were added to the cells and incubated for 24 h. MTT solution (5 mg/mL in PBS, filter-sterilized) was added to each well, and the plate was further incubated for 4 h for formazan crystal formation. The medium was carefully removed, and DMSO (100 µL) was added to each well to dissolve the formazan crystals. The plate was read using an ELISA microplate reader (EMax®, Molecular Device, Inc., USA) at an absorbance of 540 nm, with a reference wavelength of 650 nm. The cell viability, expressed as a percentage, was defined as the ratio of absorbance of treated cells to untreated cells. The 50% inhibitory concentration (IC_{50}) was interpolated from the response curve using IDBS XLFit® Software.

D. Neurite Outgrowth Stimulation Assay

NG108-15 cells were plated into 6-well poly-D-lysine (Chemicon International Inc, MA, USA)-coated cell culture plates (Orange-Scientific) at a cell density of 8×10^3 cells per well. The mushroom extract and NGF (Sigma-Aldrich) were added to the wells either individually or as combined concentrations of extract. NGF was tested in the concentration range of 5–100 ng/mL, whereas the mushroom extract at 1–500 µg/mL. For the synergistic effect of neurite stimulation, the mushroom extract (0–100 µg/mL) in combination with NGF (5, 10, 20 ng/mL) was tested. Control wells contained only cells with medium. The assay plates were incubated for 48 h at 37°C in a 5% CO_2 humidified incubator. The cells were then observed under Nikon Eclipse TS100 microscope and images were captured with Nikon's Imaging Software, NIS-Elements. The scoring of neurite outgrowth was performed as described by Wong.[14] The mean differentiation score was obtained for at least 300 cells in each well.

E. Measurement of NGF Levels Using ELISA

NG108-15 cells were seeded in 96-well plates (Orange-Scientific) at a seeding density of 1×10^4 cells per well. The cells were treated with mushroom extract, or in combination with NGF, for 48 h. Cell culture supernatant was collected, centrifuged at 1500 ×*g* for 15 min, and maintained at 0–4°C prior to assay. The samples were diluted with the provided buffer, if required. The NGF levels in cell supernatant were measured using NGF E_{max}® ImmunoAssay System (Promega Corporation, Madison, WI, USA) according to the manufacturer's protocol.

F. Neurofilament-200 Staining

NG108-15 cells were grown on poly-D-lysine coated cover slips (12 mm) and treated with mushroom extract and NGF as described in section D above. Upon incubation, the cells were fixed with 4% paraformaldehyde followed by 1.0% Triton X-100. The cells were incubated with primary antibody, anti-neurofilament 200 (Sigma-Aldrich, St Louis, MO, USA) (1:200 dilution) for 1 h in a humidified chamber. The cells were washed with 2% sheep serum in PBS, followed by reaction with FITC-conjugated sheep anti-rabbit IgG antiserum (Sigma-Aldrich) (1:100 dilution) for 1 h in dark. The cell nuclei were then counterstained with ProLong® Gold antifade reagent with DAPI (P36931; Invitrogen) and mounted on glass slides for observation.

G. Assessment of Neuroprotective Activity in *H. erinaceus* Extract Using MTT Assay

The neuroprotective activity of *H. erinaceus* extract was evaluated using MTT assay mentioned in section C, with slight modifications. Briefly,

NG108-15 cells were seeded in poly-*D*-lysine coated 96-well plates at a density of 1×10^4 cells per well. The cells were treated with *H. erinaceus* extract (1–1000 µg/mL) under two treatment modes: pre-treatment and co-incubation. In the pre-treatment mode, the cells were pre-incubated with mushroom extract followed by H_2O_2 exposure (100 µM, 2 h). The mushroom extract was introduced to the cells during the H_2O_2 exposure (100 µm, 2 h) in co-incubation mode. The cells were incubated in phenol-red free medium (Hyclone® DMEM, Catalogue No.: SH30284.01) when H_2O_2 was introduced. The addition of MTT solution and subsequent steps were similar to section C.

H. Assessment of Neuroprotective Activity in *H. erinaceus* Extract Using Trypan Blue Exclusion Assay

NG108-15 cells were plated into 12-well cell-culture plates (Orange-Scientific) at a density of 5×10^4 cells per well. The cells were similarly treated with two treatment modes, pretreatment and co-incubation, as mentioned in section G. Upon incubation, the cells were mixed with 0.4% trypan blue (ratio 1:1) and examined under microscopy using a hemocytometer. Viable healthy cells appear as clear white disks that have excluded trypan blue (unstained cells). Early apoptotic cells exclude trypan blue but with irregular shaped or shrunken nuclei. Cells in end-stage apoptosis or necrosis appear as irregular, blue-stained cells or as remnants of dead cells. A minimum of 200 total cells were counted, and the percentage of viable cells were calculated as follows:

$$\% \text{ viable cells} = \frac{\text{Total no. of live cells}}{\text{Total no. of cells counted}} \times 100.$$

I. Assessment of Neuroprotective Activity in *H. erinaceus* Extract Using a TUNEL Assay

NG108-15 cells were plated in Lab-Tek™ II – CC^{2}™ 8-well chamber slide (Nalge Nunc International, Naperville, IL, USA) at a density of 2×10^4 cells per well and treated with mushroom extract (24 h), H_2O_2 (2 h), or both in succession. A TUNEL assay was performed using a TUNEL Apoptosis Detection Kit (Millipore Corporation, Temecula, CA, USA) according to the manufacturer's protocol. The cells were counterstained with ProLong® Gold antifade reagent with DAPI (P36931, Invitrogen) and mounted on glass slides for observation.

J. Statistical Analysis

The data were statistically analyzed using one-way analysis of variance (ANOVA). Significant differences between means were determined by Duncan's multiple range test. Values with $P < 0.05$ were regarded as statistically significant.

III. RESULTS

A. Assessment of Cytotoxicity in *H. erinaceus* Aqueous Extract

There was a concentration-dependent increase of viability in MRC-5 cells exposed to the extract concentrations at 0–1000 µg/mL (Figure 1). At 1000 µg/mL, cell proliferation was observed at the end of a 24-h incubation, reflected by a 37.74% increment in cellular viability which was indicated as an increase in cell number. The 50% inhibitory concentration (IC_{50}) of *H. erinaceus* extract for MRC-5 interpolated from the response curve was 34,094.57 ± 1199.8 µg/mL. The viability of NG108-15 cells was relatively constant when exposed to the mushroom extract at 0–1000 µg/mL. There was no significant statistical difference between the groups at the range of concentrations tested. The IC_{50} value of *H. erinaceus* extract for NG108-15 was 14,446.07 ± 1548.34 µg/mL.

B. Neurite Outgrowth Stimulation of *H. erinaceus* Aqueous Extract

Both NGF and *H. erinaceus* aqueous extract induced neurite outgrowth stimulation in NG108-15 cells in a dose-dependent manner (Figure 2A). In this experimental model, the optimal NGF concen-

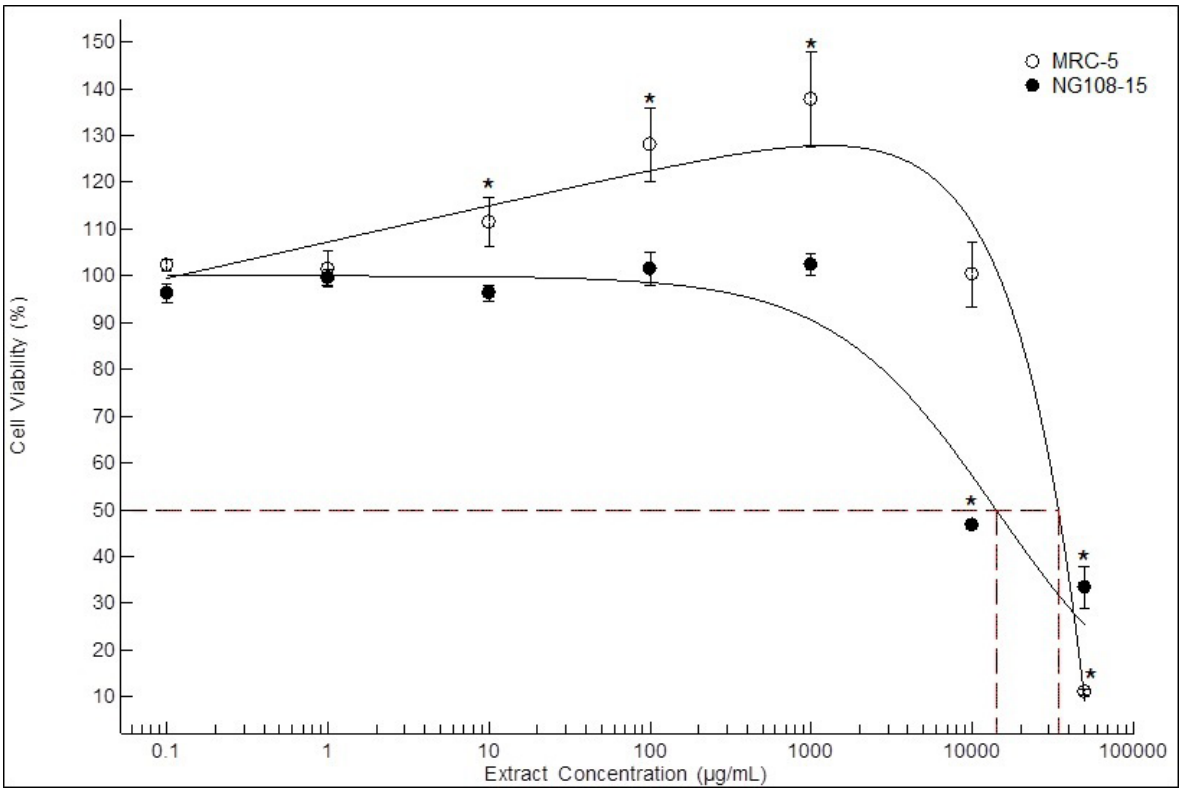

FIG. 1: Cytotoxic effect of *Hericium erinaceus* extract *in vitro*. Each data point represents the mean ± standard error of the mean (SEM) of three independent experiments carried out in triplicate. Untreated controls (0 mg/mL extract) were not plotted on log-scale graph. An asterisk (*) denotes a significant difference ($P < 0.05$) from the corresponding value for control. The 50% inhibitory concentration for the endpoint measured (IC_{50}) was interpolated from the response curve using IDBS XLfit® Software.

tration for neurite stimulation was 20 ng/mL (33.7 ± 2.9%), whereas *H. erinaceus* extract was optimally effective at 50 µg/mL (36.5 ± 2.2%). Higher concentrations of NGF or *H. erinaceus* extract did not augment the neurite outgrowth but reduced the proportion of neurite-bearing cells. NG108-15 cells were more responsive to *H. erinaceus* extract in expressing their neurite outgrowth compared to NGF treatments. The percentage of increase was higher in cells treated with optimal concentration of *H. erinaceus* extract at 50 µg/mL (64.8% increment) compared to NGF at 20 ng/mL (43.4% increment). Extracellular NGF levels in NG108-15 were significantly increased when treated with *H. erinaceus* extract in the concentration range of 1–100 µg/mL (Figure 3A). Maximum NGF secretion (45.67 ± 0.79 pg/mL) was obtained when the optimal concentration of *H. erinaceus* extract for neurite outgrowth activity (50 µg/mL) was applied.

The *H. erinaceus* extract was further studied to investigate whether the extract would enhance the activity of NGF in stimulating neurite outgrowth. NG108-15 cells were subjected to 5, 10, and 20 ng/mL NGF combined with different concentrations of *H. erinaceus* aqueous extract (0–100 µg/mL) (Figure 2B). The concentration of *H. erinaceus* extract at 500 µg/mL was not further evaluated due to reduced neurite stimulation activity at this concentration. The application of 5 ng/mL NGF induced 25.5 ± 1.9% cells to extend neurites. There was a small magnitude of increase (4.9–9.5%) in the percentage of neurite-bearing cells in the combined treatments of *H. erinaceus* extract and 5 ng/mL NGF. A considerable enhancement of

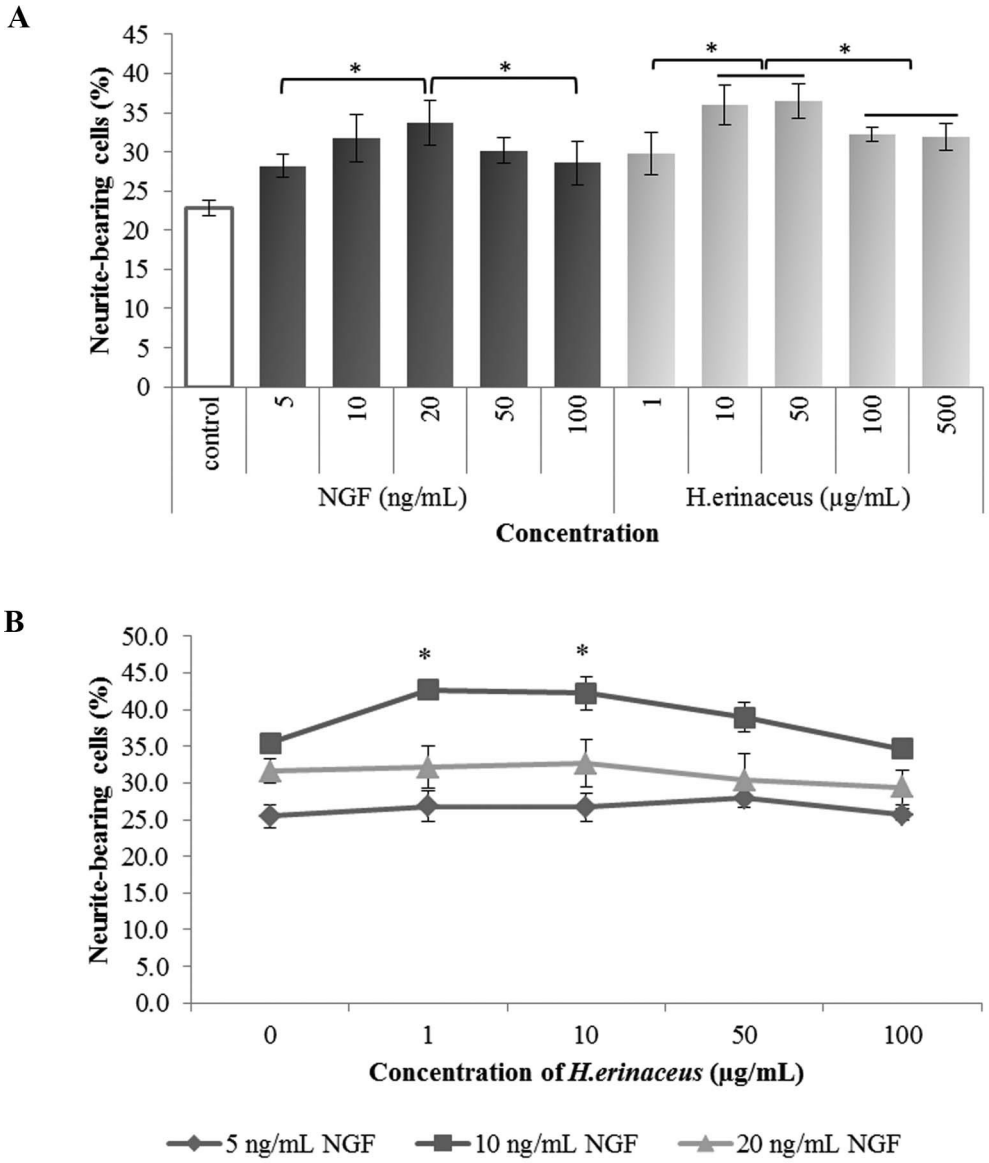

FIG. 2: Neurite outgrowth stimulation activity in NG108-15 cells treated with NGF and *Hericium erinaceus* extract (A) individually or (B) a combination of both. (A) Histogram showing the mean ± S.E.M. percentage of neurite-bearing cells treated with different concentrations of NGF and *H. erinaceus* aqueous extract for 48 h. (B) Line graph showing the mean ± S.E.M. percentage of neurite-bearing cells treated with different concentrations of NGF combined with various concentrations of *H. erinaceus* aqueous extract for 48 h. All treatment groups were statistically different ($p < 0.05$) from the untreated control. Each bar represents data from three independent experiments carried out in duplicate. An asterisk denotes a significant difference ($p < 0.05$) between the marked treatment groups.

neurite outgrowth was observed in the combined treatments of 10 ng/mL NGF and 1–10 µg/mL *H. erinaceus* extract. The combination of 10 ng/mL NGF with 1 µg/mL mushroom extract markedly enhanced the neurite outgrowth activity, with 42.7 ± 2.3% cells showing neurite extensions. These result invariably indicate an increase of 20.4% in neurite outgrowth activity compared to 10 ng/mL

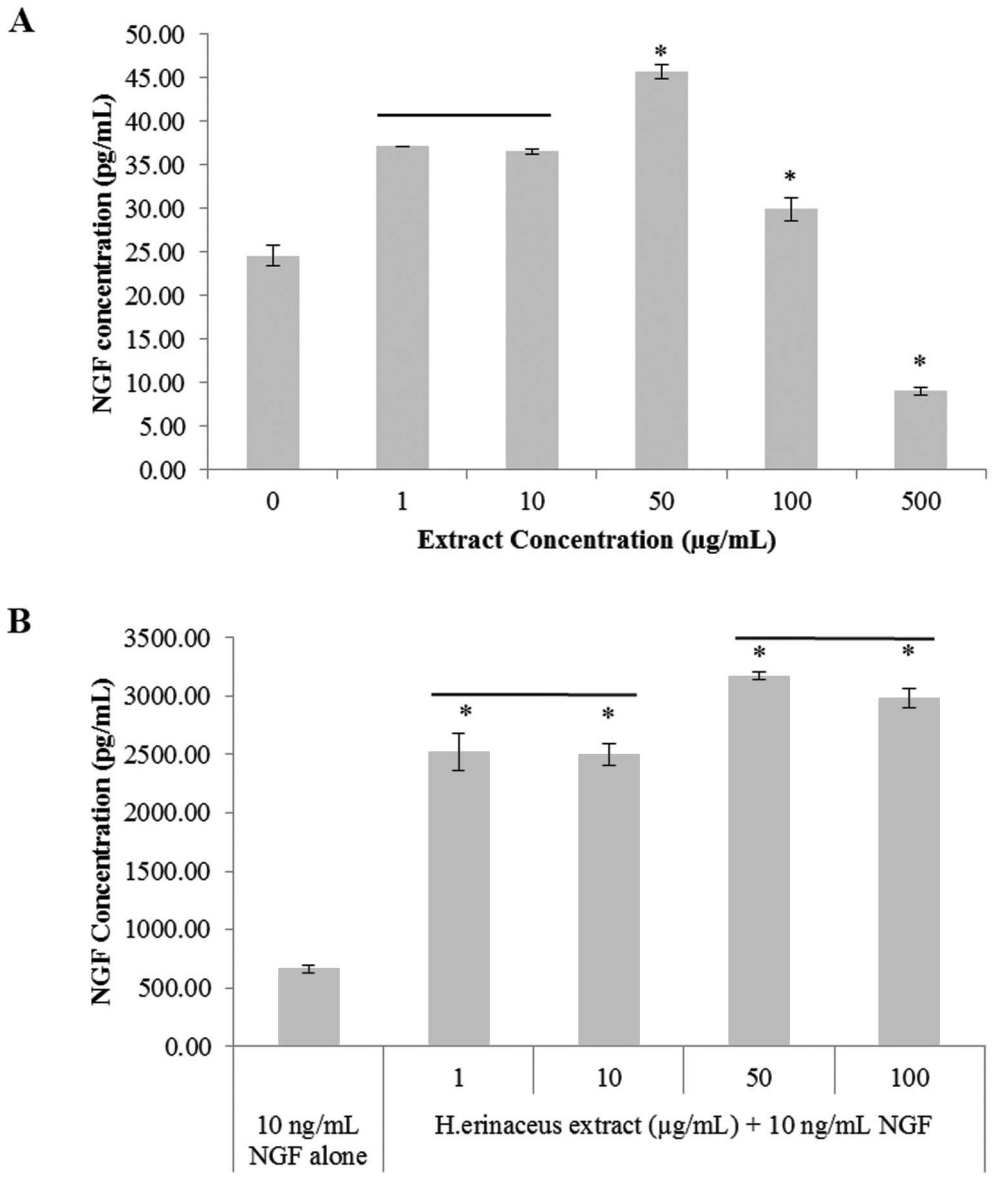

FIG. 3: Extracellular NGF concentration in NG108-15 cells upon different treatments. (A) Histogram showing the mean ± S.E.M. concentration of NGF in cells treated with *H. erinaceus* aqueous extract. All treatment groups were statistically different ($p < 0.05$) compared to the untreated control. An asterisk denotes a significant difference ($p < 0.05$) compared to cells treated with lowest concentration of extract (1 µg/mL). (B) Histogram showing the mean ± S.E.M. concentration of extracellular NGF in cells treated with *H. erinaceus* aqueous extract combined with 10 ng/mL NGF. An asterisk denotes a significant difference ($p < 0.05$) compared to 10 ng/mL NGF treatment alone. There is no significant difference between treatment groups under the horizontal line.

NGF alone. A comparable increment was also observed in the treatment of 10 ng/mL NGF together with 10 µg/mL mushroom extract of which 42.3 ± 1.9% neurite-bearing cells were recorded. The effect of neurite outgrowth was enhanced in these combined mixtures when compared to the concentration of mushroom extract and NGF applied individually. The stimulation of neurite outgrowth was

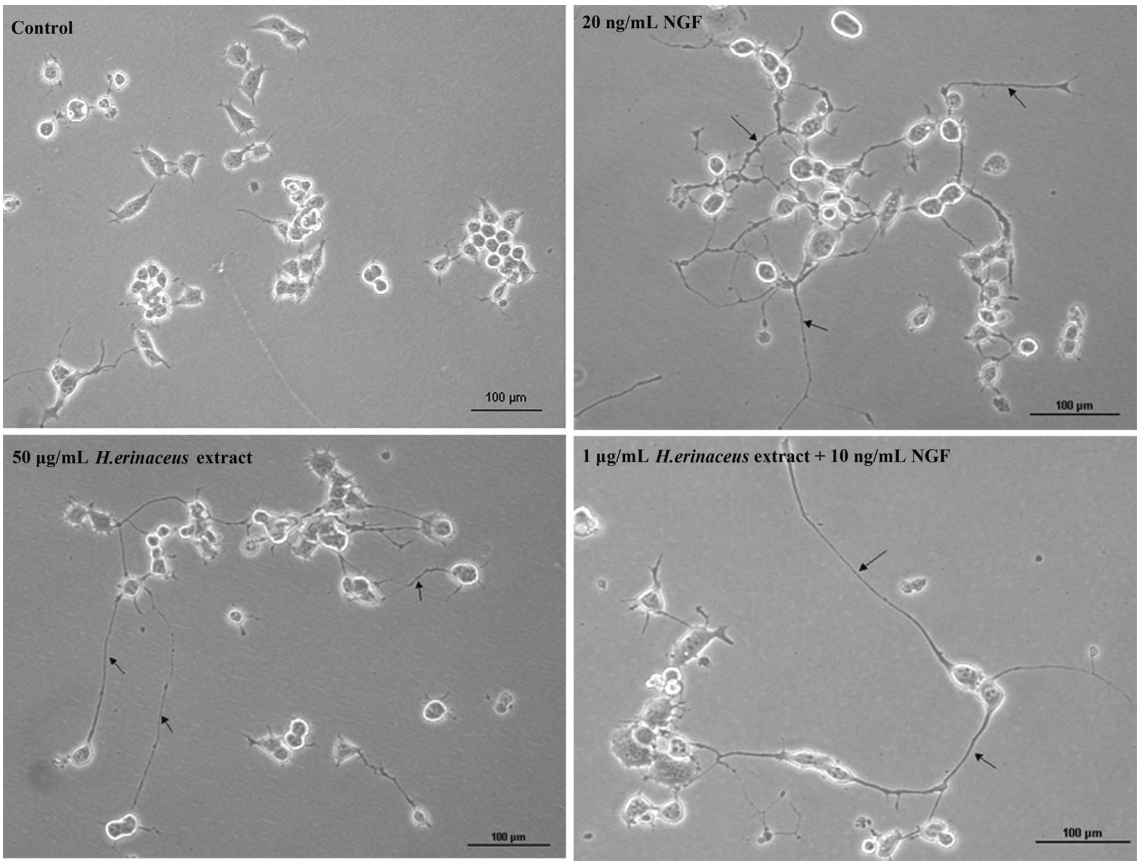

FIG. 4: Morphology of neurite-bearing cells in NG108-15 cells. Untreated control cells remained round and undifferentiated. Cells treated with NGF, *Hericium erinaceus* extract or a combination of both exhibited visible neurite extensions. Scale bar = 100 µm

enhanced, albeit to a lesser degree (9.9%), when *H. erinaceus* extract was increased to 50 µg/mL in combinatorial treatments of 10 ng/mL NGF. The enhancement of neurite outgrowth activity was not observed when the concentration of NGF was increased to 20 ng/mL in the combinatorial treatments.

The extracellular NGF in the combinatorial treatments of 10 ng/mL NGF was measured to determine whether the enhancement of neurite outgrowth was related to secreted NGF levels as in *H. erinaceus* extract-treated cells (Figure 3B). There was a stepwise increase in the extracellular NGF level in these combinatorial treatments. The concentration of extracellular NGF increased 4-fold (treated with 1–10 µg/mL *H. erinaceus* extract) and 5-fold (treated with 50–100 µg/mL extract) compared to individual application of 10 ng/mL exogenous NGF. However, the neurite outgrowth activity showed the opposite trend upon application of 10 ng/mL exogenous NGF combined with high concentrations of mushroom extract (50–100 µg/mL) with declining percentage of neurite-bearing cells.

Untreated control cells remained undifferentiated throughout the 48-h incubation period, whereby the cells were flat, round, and lacked neurite extensions. Cells treated with *H. erinaceus* extract, NGF, or combination of both exhibited neuron-like morphology with extensive cellular processes (Figure 4). Both undifferentiated and differentiated cells stained positive for neurofilament-200, indi-

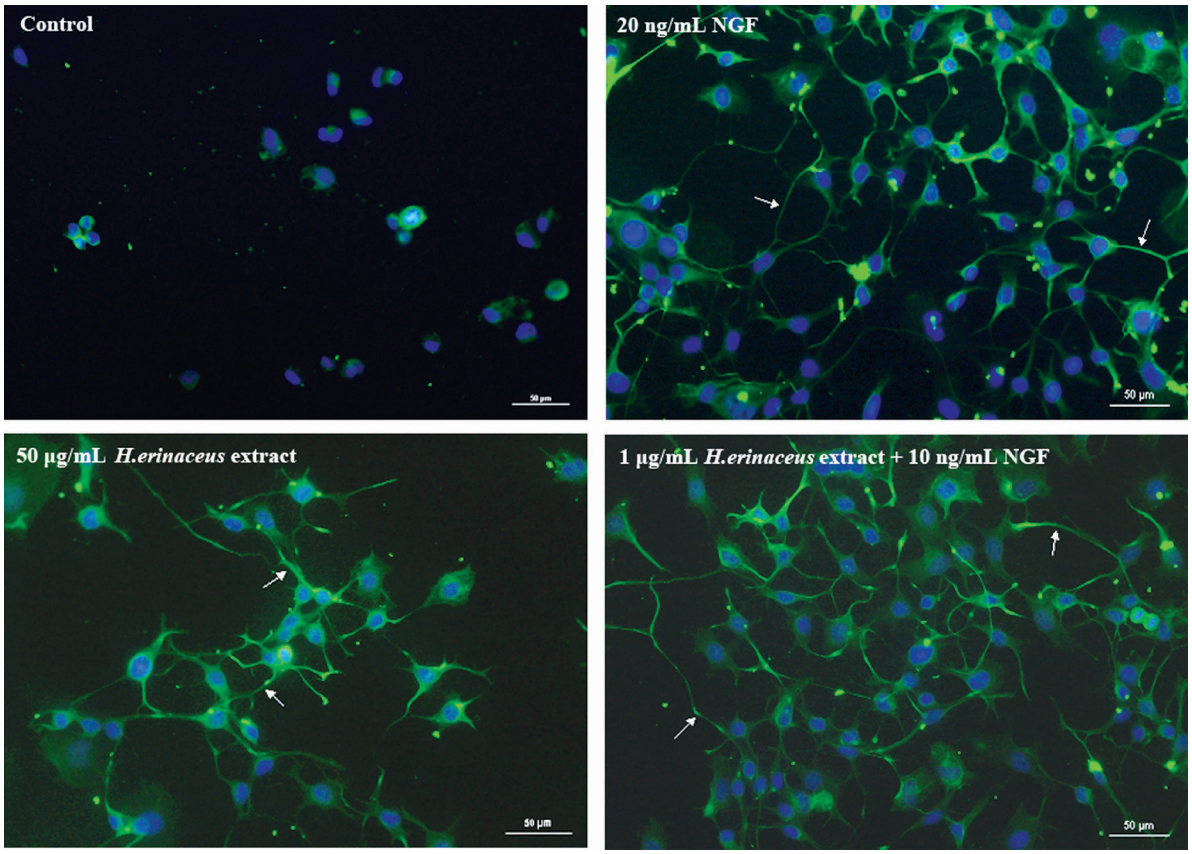

FIG. 5: Immunocytochemical staining of NG108-15 cells for neurofilament-200. The neural cells were labeled with antibodies against neurofilament-200 followed by FITC (shown in green). Cell nuclei were counterstained with DAPI (blue) to facilitate cell counting. The neurites stained positive for neurofilament-200 (white arrows) after treated with *Hericium erinaceus* extract and/or NGF. Scale bar = 50 µm.

cating that the visible neurite extensions were of neuronal origin (Figure 5).

C. Neuroprotective Effect of *H. erinaceus* Aqueous Extract

The neuronal cells were subjected to H_2O_2–induced oxidative stress to evaluate the neuroprotective effects of the aqueous extract of *H. erinaceus*. A cytotoxicity assessment of H_2O_2 toward NG108-15 cells was conducted, and the treatment condition of 100 µM H_2O_2 (2 h incubation), which results in a 50% change in viability, was chosen as the benchmark. The neuroprotective effect of *H. erinaceus* extract against oxidative stress was tested in two different modes: pre-treatment and co-treatment of extract application. In the pre-treatment mode, cells were incubated with the mushroom extract for a defined period prior to introduction of oxidative stress. In the co-treatment mode, the mushroom extract was added simultaneously with H_2O_2 to NG108-15 cells and incubated for 2 h.

The neuroprotective effect of *H. erinaceus* extract was evaluated using MTT assay which determines viability by assessing cellular metabolic activity (Figure 6A). The pre-treatment of *H. erinaceus* extract for 2 h did not improve the viability of NG108-15; there was no significant difference ($p < 0.05$) between the extract-treated and non–extract-treated cells. The cellular viability was not improved, even when the pre-incubation period was extended to 24 h. The co-incubation of *H. eri-*

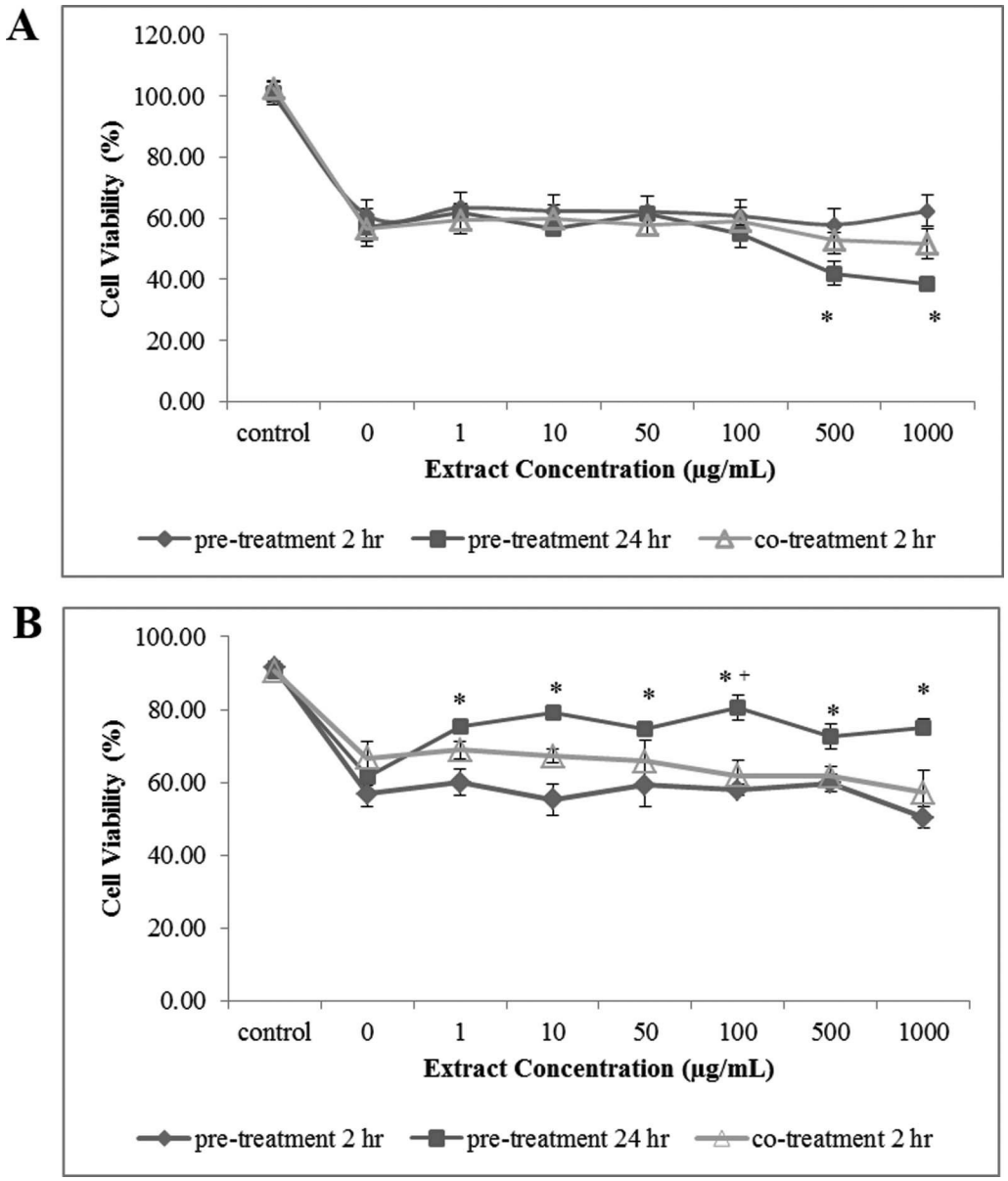

FIG. 6: Neuroprotective effects of *Hercium erinaceus* aqueous extract evaluated using (A) MTT assay and (B) trypan blue exlusion assay. Line graph showing the mean ± S.E.M. cell viability of NG108-15 cells treated with pretreatment and cotreatment modes of extract application followed by H_2O_2 incubation (2 h). Control cells were cultured in medium without the application of *H. erinaceus* extract or H_2O_2. All treatment groups were significantly different ($p < 0.05$) from untreated control. An asterisk denotes a difference ($p < 0.05$) from the 0 µg/mL condition (cells subjected to H_2O_2 without addition of extract).

naceus extract and H_2O_2 similarly did not improve the viability of NG108-15 cells.

The neuroprotective effect of *H. erinaceus* extract was also evaluated using a trypan blue exclusion assay, which determines viability by assessing membrane integrity (Figure 6B). The cellular viability was not improved when NG108-15 cells were pre-treated with *H. erinaceus* extract for 2 h.

However, an extended period of incubation of the mushroom extract did confer protection in NG108-15 cells against oxidative stress. The viability was increased to 80.52 ± 3.39% when the cells were pre-treated with 100 µg/mL *H. erinaceus* extract for 24 h. The neuroprotective effect was not observed when the same concentration of *H. erinaceus* extract was applied in the co-treatment mode.

As the pre-treatment of *H. erinaceus* extract at 100 µg/mL (24 h incubation) prior H_2O_2 exposure showed significant protection when assessed using trypan blue exclusion assay, this treatment condition was further evaluated using a TUNEL assay which identifies the presence of apoptotic cells (Figure 7). A fraction of apoptotic cells (17.68 ± 3.41 %) was detected when the cells were incubated with 100 µM H_2O_2 for 2 h. Pre-incubation of *H. erinaceus* extract (24 h incubation) slightly reduced the apoptotic cells to 15.85 ± 3.05 %. However, this reduction was not significantly different ($p < 0.05$) when compared with H_2O_2 treatment alone.

IV. DISCUSSION

MTT is a rapid spectrophotometric assay for assessing cellular viability, whereby the yellow tetrazolium salt is reduced to insoluble purple formazan crystals by the mitochondria of viable cells.[18] A close correlation between MTT assay and cell resting membrane potential (RMP) measurements, a sensitive health indicator for nerve cells, denotes the suitability of MTT assay in evaluating cytotoxicity in neuronal cells *in vitro*.[19]

The MRC-5 is a cell line derived from normal human lung tissue commonly used for *in vitro* cytotoxicity testing. Cytotoxic phenols isolated from *H. erinaceus*, hericenone A and B, exhibited cytotoxicity toward cervical cancer cell lines (HeLa).[20] Yet, the cytotoxicity of *H. erinaceus* has not been tested on other cell lines. Cell proliferation was observed in MRC-5 cells treated with 10–1000 µg/mL *H. erinaceus* extract. Fibroblast growth and proliferation are critical to wound healing with the production of growth factors for intercellular signalling and proteins for remodelling of the clot and extracellular matrix.[21] Consistent with this, the topical application of aqueous extract of locally grown *H. erinaceus* was reported to enhance wound healing enclosure in rats.[22]

Conversely, cell proliferation was not observed in NG108-15 cells treated with 0.1–1000 µg/mL *H. erinaceus* extract. Neuronal differentiation has been closely linked to cessation of cell proliferation. The induction of nitric oxide synthase (NOS) plays an important role in cellular growth arrest, initiating the switch to cytostasis during differentiation.[23] The rate of cell proliferation decreased when NG108-15 cells were incubated with differentiation medium containing dibutyryl cAMP.[24] Hence, the absence of cell proliferation in NG108-15 when incubated with *H. erinaceus* extract may be attributed to neuronal differentiation. The IC_{50} values of *H. erinaceus* in both cell lines were relatively high in comparison to the IC_{50} value of known cytotoxic agents, which is usually less than 1000 µg/ml (1 mg/mL). This denotes that the *H. erinaceus* extract had no cytotoxic effects on MRC-5 and NG108-15 cell lines.

NG108-15 cells are round, tend to pile up and grow in clusters in nature. In this study, the poly-D-lysine coating served as a neutral substratum where the cells attached within the first hour of plating, and this minimized cells clustering. Upon incubation, untreated neuronal cells had short cellular extensions, but most did not reach the criterion length of one cell body diameter to be scored as a neurite-bearing cell. NG108-15 cells treated with NGF and/or *H. erinaceus* extracts exhibit multipolar, highly branched, beaded extensions expressing 'rapid-onset' neurites. Nonetheless, the presence of 'slow-onset' neurites of unipolar, smooth, and cable-like extensions was similarly observed. This reflects the presence of two different types of neurite patterns within the population after 48 h of incubation.[25]

Neurofilaments are one of the major neuronal protein building blocks in the cytoskeleton of neurites. These neuronal proteins are useful immunocytochemical markers for axons due to their abundance in mature axons.[26] Neurite extensions

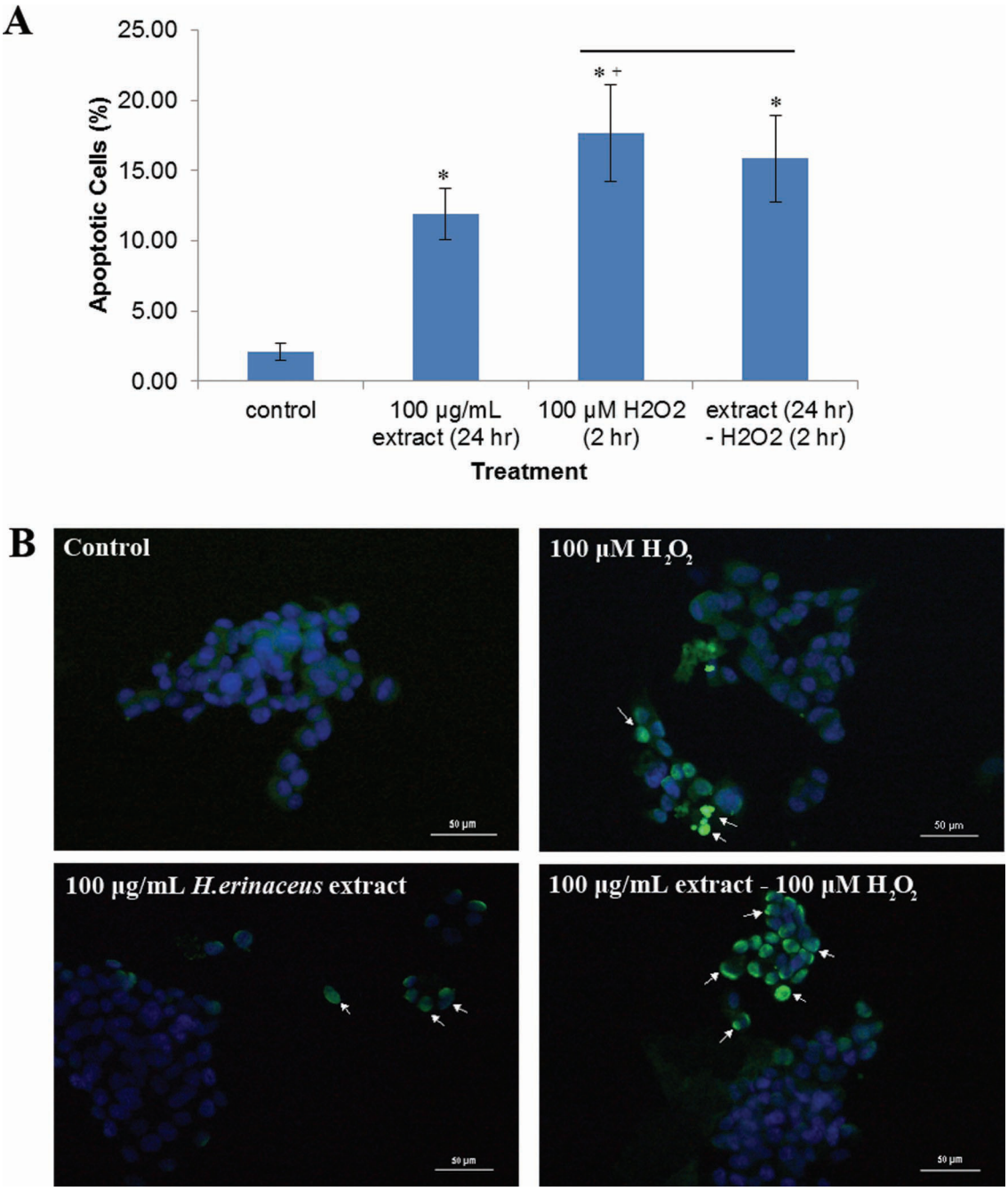

FIG. 7: Apoptotic cells subjected to different treatments evaluated using TUNEL assay. (A) Histogram showing the mean ± S.E.M. percentage of apoptotic cells treated with *Hericium erinaceus* extract, H_2O_2 or both. An asterisk denotes significant difference compared to untreated control whereas a plus sign denotes significant difference compared to 100 µg/mL *H. erinaceus* extract (24 h incubation). There is no significant difference between the treatment groups under the horizontal line. ($p < 0.05$). (B) Morphology of apoptotic cells stained using TUNEL assay. Apoptotic cells were stained green (avidin-FITC) and display morphological properties such as cellular condensation and fragmentation. Scale bar = 50 µm.

of cultured cells may be related to outgrowth of axonal or dendritic processes. The NG108-15 cell line has been reported to extend axon-like and dendrite-like processes when induced with different neuritogenic agents.[27] The positive staining of neurofilament-200 in NG108-15 cells treated with *H. erinaceus* extract and/or NGF may indicate the formation of axon-like outgrowth.

The treatment of *H. erinaceus* aqueous extract, applied individually or in combination with NGF, induced visible neurite extensions in NG108-15 cells compared to untreated control cells. The cells appeared to be more sensitive and responsive to the mushroom extract (36.5 ± 2.2% neurite-bearing cells) compared to NGF (33.7 ± 2.9% neurite-bearing cells) with respect to neurite outgrowth stimulation. Extracellular NGF levels in NG108-15 were profoundly increased when treated with *H. erinaceus* extract. A similar bell-shaped trend occurred in the neurite outgrowth and extracellular NGF levels where both activity expressions were optimal at 50 µg/mL extract treatment. This reflects an association between secreted NGF levels in culture and neurite outgrowth activity in NG108-15 cells treated with *H. erinaceus* extract. The neuroactive compounds in the aqueous preparation act as inducers of NGF-synthesis in the neuronal cells rather than having neuritogenic activity *per se*.

The combined treatments of *H. erinaceus* extract and 10 ng/mL NGF enhanced the neurite outgrowth activity of NGF (Figure 2B). The individual application of 10 ng/mL NGF and 1 µg/mL *H. erinaceus* extract yielded 33.3% and 34.5% increments compared with the control, respectively. The combination of these concentrations showed an additive response, with 60.6% increment, which was attributed to elevated levels of extracellular NGF in culture. However, the additive effect was not observed with increasing concentrations of *H. erinaceus* extract and appeared to be saturated at 1 µg/mL. The application of 10 ng/mL NGF combined with high concentrations of *H. erinaceus* extract (50 – 100 µg/mL) increased extracellular NGF levels but reduced the proportion of neurite-bearing cells. A high level of extracellular NGF may have inhibited the neurite outgrowth activity in NG108-15 cells. The signalling of Trk receptor by neurotrophins can adversely affect neuronal survival under certain circumstances.[28]

Both MTT and trypan blue exclusion assays are regular cell-viability assays whereby MTT evaluates cellular metabolic activity and the latter evaluates membrane structural integrity. Although the pre-treatment of *H. erinaceus* aqueous extract (24 h incubation) demonstrated protective effects against oxidative stress when assessed using trypan blue exclusion assay, the disruption of metabolic activity in NG108-15 cells was clearly revealed by MTT assay. During oxidative stress, mitochondrial membrane transition pore permeability is increased, followed by loss of mitochondrial NAD^+ and further generation of superoxide radicals leading to cell injury. The presence of free radicals impairs mitochondrial electron transport which leads to mitochondrial dysfunction.[29] On the contrary, cell membrane of cells undergoing apoptosis remains intact for a relatively long time; only cells in late stages of apoptosis and necrotic cells take up the dye trypan blue and appear as blue cells.[30]

The TUNEL assay was employed to further clarify the obscure protective effect of the mushroom extract. The TUNEL method is based on direct, specific, *in situ* labelling of DNA fragmentation sites in the nuclei of fixed cells. Apoptotic cells exhibit distinct morphological hallmarks, including nuclear condensation and formation of pyknotic bodies of condensed chromatin.[30] It was observed that cells treated with *H. erinaceus* extract prior to being subjected to oxidative stress showed similar apoptotic features present in non-extract treated cells. Moreover, there was no significant difference between the percentages of apoptotic cells present in both treatment groups. This result confirms the absence of a protective effect in the mushroom aqueous extract.

The aqueous extract of locally grown *H. erinaceus* was reported to possess antioxidant properties with the presence of high phenolic content.[31] However, the moderate antioxidant level was insufficient to counteract oxidative insults and confer protection to NG108-15 cells under oxidative

stress, as shown in this study. Nonetheless, enzymatic extracts of *H. erinaceus* showed more effective antioxidative and superoxide radical scavenging-activity compared to water and organic solvent extracts of the mushroom. Pepsin-treated extracts exhibited neuroprotective effects against H_2O_2-induced oxidative stress in PC12 cells by regulating the anti-apoptotic protein Bcl-2[32]. Hence, different extract preparations of locally grown *H. erinaceus* must be further studied for their neuroprotective effects.

Neuroprotective molecules were extracted from *H. erinaceus,* including dilinoleoyl-phosphatidylethanolamine (DLPE),[9] 3-hydroxyhericenone F[33] and several unnamed compounds from the scrap mushroom cultivation bed.[34] A patent[35] was filed to extract an anti-dementia substance from a fat-soluble fraction which contains benzyl alcohol derivatives, chromane derivatives, and phosphatidyethanolamine derivatives as main bioactive compounds. The bioactive fraction was reported to increase the synthesis of NGF and to reduce toxicity of amyloid-β peptide. These studies indicated that the neuroprotective compounds present in *H. erinaceus* are of lower polarity and can be mainly extracted with organic solvents. Hence, these neuroprotective compounds are most likely absent in the aqueous preparation of *H. erinaceus* used in this study.

Several sugars were isolated from the hot water preparation of locally grown *H. erinaceus,* with arabinose being the major sugar component.[36] Arabinose was not detected in the chemical profile of *H. erinaceus* grown in China. An exobiopolymer purified from the liquid culture broth of *H. erinaceus* mycelium enhanced the growth and neurite extension of PC12 cells.[37] Thus, there may be bioactive polysaccharides in the aqueous extract of *H. erinaceus* responsible for the neurite outgrowth activity in NG108-15 cells observed in this study. The phytochemical screening of chemical constituents of *H. erinaceus* aqueous extract used in this study showed that the extract was rich in phenolics and triterpenoids (unpublished data). It is worthwhile to further investigate the neuroactive components in the aqueous extract of locally grown *H. erinaceus*.

The study of neurotrophic factors as therapeutic agents in neurodegenerative diseases has long been contemplated. It is known that the balance of NGF trophic system is altered in the brains of Alzheimer's disease patients. Thus, with NGF acting as a therapeutic agent, we can deduce that this neurotrophic factor can impede further death of cholinergic neurons and can restore functions of degenerating cells.[38] Current drug development studies have focused on the therapeutic potential of CERE-110 (AAV2-NGF), an adeno-associated virus-based gene delivery vector that encodes human NGF.[39] CERE-110 has successfully passed phase 1 clinical trials, and phase 2 clinical testing is currently being conducted.[40] As the aqueous *H. erinaceus* extract was shown to enhance neurite differentiation when used in combination with NGF, the supplementation of this mushroom extract in NGF therapeutic treatments may be beneficial. However, further studies must be conducted to investigate whether the enhancement effect of neurite outgrowth activity by NGF combined with *H. erinaceus* aqueous extract is effective *in vivo*.

ACKNOWLEDGMENTS

The authors would like to gratefully acknowledge the financial support provided by MOSTI, Malaysia (SF12-02-03-2050) and University Malaya (PS285/2008C and RP005B-2013AFR).

REFERENCES

1. Auld DS, Kornecook TJ, Bastianetto S, Quirion R. Alzheimer's disease and the basal forebrain cholinergic system: Relations to β-amyloid peptides, cognition, and treatment strategies. Prog Neurobiol. 2002;68(3):209–45.
2. Mizuno T. Bioactive substances in Hericium erinaceus (Bull.: Fr.) Pers.(Yamabushitake), and its medicinal utilization. Int J Med Mushrooms. 1999;1:105–19.
3. Kawagishi H, Ando M, Sakamoto H, Yoshida S, Ojima F, Ishiguro Y, Ukai N, Furukawa S. Hericenones C, D and E, stimulators of nerve growth factor (NGF)-synthesis, from the mushroom Hericium erinaceum. Tetrahedron Lett. 1991;32(35):4561–64.
4. Kawagishi H, Ando M, Shinba K, Sakamoto H, Yoshida S, Ojima F, Ishiguro Y, Ukai N, Furukawa S. Chromans, hericenones F, G and H from the mushroom Hericium

erinaceum. Phytochemistry. 1993;32(1):175–78.
5. Kawagishi H, Shimada A, Shirai R, Okamoto K, Ojima F, Sakamoto H, Ishiguro Y, Furukawa S. Erinacines A, B and C, strong stimulators of nerve growth factor (NGF)-synthesis, from the mycelia of Hericium erinaceum. Tetrahedron Lett. 1994;35(10):1569–72.
6. Kawagishi H, Shimada A, Shizuki K, Mori H, Okamoto K, Sakamoto H, Furukawa S. Erinacine D, a stimulator of NGF-synthesis, from the mycelia of Hericium erinaceum. Heterocyclic Communications. 1996;2(1):51–54.
7. Kawagishi H, Shimada A, Hosokawa S, Mori H, Sakamoto H, Ishiguro Y, Sakemi S, Bordner J, Kojiman N, Furukawa S. Erinacines E, F, and G, stimulators of nerve growth factor (NGF)-synthesis, from the mycelia of Hericium erinaceum. Tetrahedron Lett. 1996;37(41):7399–402.
8. Lee EW, Shizuki K, Hosokawa S, Suzuki M, Suganuma H, Inakuma T, Li J, Ohnishi-Kameyama M, Nagata T, Furukawa S, Kawagish H. Two novel diterpenoids, erinacines H and I from the mycelia of Hericium erinaceum. Biosci Biotechnol Biochem. 2000;64(11):2402–5.
9. Nagai K, Chiba A, Nishino T, Kubota T, Kawagishi H. Dilinoleoyl-phosphatidylethanolamine from Hericium erinaceum protects against ER stress-dependent Neuro2a cell death via protein kinase C pathway. J Nutr Biochem. 2006;17(8):525–30.
10. Mori K, Inatomi S, Ouchi K, Azumi Y, Tuchida T. Improving effects of the mushroom Yamabushitake (Hericium erinaceus) on mild cognitive impairment: A double-blind placebo-controlled clinical trial. Phytother Res. 2009;23(3):367–72.
11. Cao X, Shoichet MS. Investigating the synergistic effect of combined neurotrophic factor concentration gradients to guide axonal growth. Neurosci. 2003;122(2):381–89.
12. Logan A, Ahmed Z, Baird A, Gonzalez AM, Berry M. Neurotrophic factor synergy is required for neuronal survival and disinhibited axon regeneration after CNS injury. Brain. 2006;129(2):490–502.
13. Madduri S, Papaloïzos M, Gander B. Synergistic effect of GDNF and NGF on axonal branching and elongation in vitro. Neurosci Res. 2009;65(1):88–97.
14. Wong KH, Sabaratnam V, Abdullah N, Naidu M, Keynes R. Activity of aqueous extracts of lion's mane mushroom Hericium erinaceus (Bull.: Fr.) Pers. (Aphyllophoromycetideae) on the neural cell line NG108-15. Int J Med Mushrooms. 2007;9(1):57–65.
15. Wong KH, Naidu M, David RP, Abdulla MA, Abdullah N, Kuppusamy UR, Sabaratnam V. Peripheral nerve regeneration following crush injury to rat peroneal nerve by aqueous extract of medicinal mushroom Hericium erinaceus (Bull.:Fr.) Pers. (Aphyllophoromycetideae). Evid Based Complement Alternat Med. 2011 (Article ID 580752).
16. Halpern GM. Healing mushrooms: effective treatments for today's illnesses. New York: Square One Publishers; 2007.
17. American Type Culture Collection. Product information sheet for HB-12317. Cell line designation: NG108-15 [108CC15]. ATCC catalog no.: HB-12317: American Type Culture Collection.
18. Edmondson JM, Armstrong LS, Martinez AO. A rapid and simple MTT-based scpectrophotometric assay for determining drug sensitivity in monolayer cultures. J Tissue Cult Methods. 1988;11(1):15–18.
19. Xie K, Harvey AL. Evaluation of nerve cell toxicity in vitro by electrophysiological and biochemical methods. Toxicol In Vitro. 1993;7(3):275–79.
20. Kawagishi H, Ando M, Mizuno T. Hericenone A and B as cytotoxic principles from the mushroom Hericium erinaceum. Tetrahedron Lett. 1990;31(3):373–76.
21. Mariggio MA, Cassano A, Vinella A, Vincenti A, Fumarulo R, Lo Muzio L, Maiorano E, Ribatti D, Favia G. Enhancement of fibroblast proliferation, collagen biosynthesis and production of growth factors as a result of combining sodium hyaluronate and aminoacids. Int J Immunopathol Pharmacol. 2009;22(2):485–92.
22. Abdulla MA, Fard AA, Sabaratnam V, Wong KH, Kuppusamy UR, Abdullah N, Ismail S. Potential activity of aqueous extract of culinary-medicinal lion's mane mushroom, Hericium erinaceus (Bull.: Fr.) Pers. (Aphyllophoromycetideae) in accelerating wound healing in rats. Int J Med Mushrooms. 2011;13(1):33–39.
23. Peunova N, Enikolopov G. Nitric oxide triggers a switch to growth arrest during differentiation of neuronal cells. Nature. 1995;375:68–73.
24. Hamprecht B. Structural, electrophysiological, biochemical, and pharmacological properties of neuroblastoma-glioma cell hybrids in cell culture. Int Rev Cytol. 1977;49:99–170.
25. Smalheiser NR. Analysis of slow-onset neurite formation in NG108-15 cells: Implications for a unified model of neurite elongation. Dev Brain Res. 1989;45(1):49–57.
26. Audesirk G, Audesirk T. Neurite development. In: Slikker WJ, Chang LW, eds. Handbook of developmental neurotoxicology. San Diego: Academic Press; 1998:61–86.
27. Wu G, Fang Y, Lu ZH, Ledeen RW. Induction of axon-like and dendrite-like processes in neuroblastoma cells. J Neurocytol. 1998;27(1):1–14.
28. Kalb R. The protean actions of neurotrophins and their receptors on the life and death of neurons. Trends Neurosci. 2005;28(1):5–11.
29. Maiese K, Chong ZZ, Hou J, Shang YC. Oxidative stress: biomarkers and novel therapeutic pathways. Exp Gerontol. 2010;45:217–34.
30. Zhivotovsky B, Orrenius S. Assessment of apoptosis and necrosis by DNA fragmentation and morphological criteria. Cellular Aging and Death. 2001;Unit 18.3(Supplement 12):18.3.1–.3.23.
31. Abdullah N, Ismail SM, Aminudin N, Shuib AS, Lau BF. Evaluation of selected culinary-medicinal mushrooms

for antioxidant and ACE inhibitory activities. Evid Based Complement Alternat Med. 2012 (Article ID 464238).
32. Lee SJ, Kim EK, Hwang JW, Kim C, Choi D, Lim B, Moon SH, Jeon BT, Park PJ. Neuroprotective effect of Hericium erinaceum against oxidative stress on PC12 cells. J Korean Soc Appl Biol Chem. 2010;53(3):283–89.
33. Ueda K, Tsujimori M, Kodani S, Chiba A, Kubo M, Masuno K, Sekiya A, Nagai K, Kawagishi H. An endoplasmic reticulum (ER) stress-suppressive compound and its analogues from the mushroom Hericium erinaceum. Bioorg Med Chem. 2008;16(21):9467–70.
34. Ueda K, Kodani S, Kubo M, Masuno K, Sekiya A, Nagai K, Kawagishi H. Endoplasmic reticulum (ER) stress-suppressive compounds from scrap cultivation beds of the mushroom Hericium erinaceum. Biosci Biotechnol Biochem. 2009;73(8):1908–10.
35. Zhuang C, Kawagishi H, Zhang L, Anzai H, Inventors. Anti-dementia substance from Hericium erinaceum and method of extraction. US patent 20090274720 (A1). Nov 5, 2009.
36. Choong YK, Rashid BAA, Young SI, Ismail Z. Quantification and identification of polysaccharide contents in Hericium erinaceus. Nutrition Food Sci. 2007;37(4):260–71.
37. Park YS, Lee HS, Won MH, Lee JH, Lee SY, Lee HY. Effect of an exo-polysaccharide from the culture broth of Hericium erinaceus on enhancement of growth and differentiation of rat adrenal nerve cells. Cytotechnology. 2002;39(3):155–62.
38. Williams BJ, Eriksdotter-Jonhagen M, Granholm A-C. Nerve growth factor in treatment and pathogenesis of Alzheimer's disease. Prog Neurobiol. 2006;80(3):114–28.
39. Bishop KM, Hofer EK, Mehta A, Ramirez A, Sun L, Tuszynski M, Bartus RT. Therapeutic potential of CERE-110 (AAV2-NGF): targeted, stable, and sustained NGF delivery and trophic activity on rodent basal forebrain cholinergic neurons. Exp Neurol. 2008;211(2):574-84.
40. ClnicalTrials.gov [website]. CERE-110 in Subjects With Mild to Moderate Alzheimer's Disease [updated 2010 June 30; cited 2011 May]. Available from: http://clinicaltrials.gov/ct2/show/NCT00087789.